

AI-Powered Career & Skill Gap Analyzer

Comprehensive System Architecture, AI Methodology & Technical Documentation

GitHub Repository: <https://github.com/Shivrajpandit/AI-Powered-Career-Skill-Gap-Analyzer>

Tech Stack: FastAPI (Python 3.11+) | Sentence-Transformers | React 18 | TypeScript | Tailwind CSS | Docker Compose

Core Purpose: Explainable ATS resume scoring, semantic skill gap detection, and personalized 12-week upskilling roadmaps.

1. Executive Summary

The **AI-Powered Career & Skill Gap Analyzer** is an explainable full-stack AI platform designed to help job candidates, students, and engineers understand how closely their resume matches real-world job requirements. Traditional ATS checkers return opaque percentage scores; this platform provides transparent, actionable insights using **rule-based extraction**, a **500+ skill taxonomy**, **SBERT dense vector embeddings**, **explainable scoring**, and an **automated 12-week roadmap generator**.

2. High-Level Architecture & End-to-End Workflow

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[ Client Layer: React 18 + TypeScript + Tailwind CSS ]  
  
- Landing Page, Dashboard, Resume Upload (PDF/DOCX), Job Posting Parser  
  
- Analysis Detail (Score Dial, Radar Charts, Badge Matrix) & 12-Week Interactive Roadmap  
  
- Multi-Job Comparison Matrix (Benchmarking single candidate across multiple target roles)  
  
|  
  
v REST API Requests (Bearer JWT Authentication)  
  
[ Backend Gateway: FastAPI (Python 3.11+) at /api/v1 ]  
  
- /auth : User registration, login, JWT token issuance & profile  
  
- /resumes : Multi-format parsing (PyMuPDF / python-docx) & entity sanitization  
  
- /jobs : Requirement extraction & structured skill classification  
  
- /analysis : Hybrid semantic matching, gap classification & explainable scoring  
  
- /roadmap : 12-Week progressive milestone roadmap generation  
  
|  
  
v AI & Core Intelligence Engine  
  
- Document Parser & Heuristic Section Segmenter  
  
- Skill Knowledge Base (500+ Canonical Skills across 8 Tech Domains)  
  
- Sentence-Transformers Model (all-MiniLM-L6-v2, 384-dimensional dense embeddings)  
  
- False-Positive Guardrail (Blacklist: Java!=JS, C++!=C#, Rust!=Ruby, React!=React Native)  
  
- Resume Quality Auditor (Action verbs, metrics, contact completeness)  
  
|  
  
v Persistence Layer  
  
[ Database: PostgreSQL (Production) / SQLite (Local) via SQLAlchemy 2.0 ORM ]
```

3. Algorithmic Methodology & Scoring Formulation

The matching engine evaluates candidate competencies against role requirements via a 4-tier cascade:

- **Tier 1 (Exact & Canonical Alias Match):** Canonical mapping against taxonomy (e.g., *React.js* -> *React*, *K8s* -> *Kubernetes*, *AWS Cloud* -> *AWS*) yields a match score of **1.0**.
- **Tier 2 (False-Positive Blacklist Guardrail):** Hard-coded domain exclusions eliminate false vector overlaps between distinct technologies like (Java, JavaScript), (C, C++), (C++, C#), (Rust, Ruby), and (React, React Native), setting their score to **0.0**.
- **Tier 3 (Category Boost & SBERT Embeddings):** Unmapped terms are encoded using all-MiniLM-L6-v2 (384-d dense vectors). Shared taxonomy domain categories add a +0.50 category boost. Cosine similarity ≥ 0.72 qualifies as a semantic/partial match.

- **Tier 4 (Gap Classification):** Skills with similarity < 0.72 are marked as missing critical gaps.

Explainable ATS Weighted Scoring Breakdown:

Component	Weight	Evaluation Formula & Criteria
Hard Technical Skills	50%	Ratio of exact matches (1.0) and partial semantic matches (0.50-0.85) against required job skills.
Soft Skills	10%	Presence of leadership, problem-solving, agile communication, and teamwork keywords.
Experience Alignment	20%	Candidate total years of experience vs. target role minimum required years.
Education Relevance	10%	Degree level matching (Bachelors, Masters, PhD) and technical field alignment.
Resume Quality & Format	10%	Presence of action verbs, quantifiable metrics/percentages, clear section headings, and contact info.

4. Technical Skills Taxonomy (8 Core Domains)

Domain	Technologies & Frameworks Covered (500+ Skills)
Backend Development	Python, FastAPI, Django, Flask, Node.js, Express, Go, Java, Spring Boot, C#, .NET, Rust, Ruby on Rails, GraphQL, REST APIs, Microservices, gRPC
Frontend Development	React, Next.js, Vue, Angular, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, SASS, Redux, Webpack, Vite, Responsive Design
Cloud & DevOps	AWS (EC2, S3, Lambda, ECS), GCP, Azure, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, GitLab), Ansible, Nginx, Prometheus, Grafana
AI & Data Science	PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, Hugging Face, NLP, LLMs, Computer Vision, LangChain, RAG, MLflow, Apache Spark
Databases & Storage	PostgreSQL, MySQL, MongoDB, Redis, Cassandra, SQLite, DynamoDB, Elasticsearch, Prisma, SQLAlchemy
Mobile Development	React Native, Flutter, Swift, iOS, Kotlin, Android SDK, Dart, Jetpack Compose
Testing / QA	Pytest, Jest, Cypress, Playwright, Selenium, Unit Testing, Integration Testing, TDD
Cybersecurity	OAuth2, JWT, OWASP Top 10, Penetration Testing, Cryptography, IAM, SOC2, SSL/TLS

5. Personalized 12-Week Upskilling Roadmap Plan

Phase	Duration	Target Goals & Deliverables
Phase 1: Foundations	Weeks 1 - 3	Core language fundamentals, environment setup, syntax, and foundational missing prerequisites.
Phase 2: Core Competencies	Weeks 4 - 6	Framework mastery, database integration, REST API development, and hands-on micro-projects.
Phase 3: Advanced Topics	Weeks 7 - 9	Cloud deployment, architectural patterns, performance optimization, CI/CD, and security testing.
Phase 4: Capstone & Prep	Weeks 10 - 12	End-to-end full-stack capstone project, GitHub portfolio polish, and technical interview preparation.

6. Database Entity-Relationship (ER) Schema

Entity	Key Columns	Description & Relationships
User	id, email, hashed_password, full_name, is_active, created_at	Account authentication, bcrypt password security, owns Resumes and Jobs.
Resume	id, user_id, title, file_path, raw_extracted_data, skills, experience_years	Stores parsed document sections, extracted skills array, and candidate profile.
JobDescription	id, user_id, title, company, raw_text, required_skills, preferred_skills	Stores target role postings with mandatory and optional skill lists.
MatchAnalysis	id, resume_id, job_id, overall_score, matched_skills, missing_skills, score_breakdown	Persists evaluation metrics, matched skills, gaps, and quality suggestions.
LearningRoadmap	id, analysis_id, total_weeks, phases, weekly_schedule	Generated 12-week schedule with structured milestones and resource links.

7. REST API Endpoints Specification

Method	Endpoint	Functionality	Auth
POST	/api/v1/auth/register	Create a new user account	Public
POST	/api/v1/auth/login	OAuth2 password flow / JWT Bearer token	Public
GET	/api/v1/auth/me	Get current authenticated user profile	JWT
POST	/api/v1/resumes/upload	Upload & parse resume (PDF / DOCX)	JWT
GET	/api/v1/resumes/	List all uploaded user resumes	JWT
POST	/api/v1/jobs/	Create & parse target job description	JWT
GET	/api/v1/jobs/	List all job descriptions	JWT
POST	/api/v1/analysis/run	Execute match analysis between resume and job	JWT
GET	/api/v1/analysis/{id}	Retrieve full match analysis breakdown	JWT
GET	/api/v1/roadmap/{analysis_id}	Retrieve / generate 12-week roadmap	JWT
GET	/health	Health check & service version	Public

8. Installation & Deployment Guide

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# Option 1: Docker Compose (Full Stack Deployment)

git clone https://github.com/Shivrajpandit/AI-Powered-Career-Skill-Gap-Analyzer.git
cd AI-Powered-Career-Skill-Gap-Analyzer
docker compose up --build

# Option 2: Local Development Setup

# Backend (FastAPI):
cd backend && python -m venv .venv && .venv\Scripts\activate
pip install -r requirements.txt && uvicorn app.main:app --reload --port 8000

# Frontend (React 18 + Vite):
cd frontend && npm install && npm run dev
```